

Micro Econ Hw 1



Q1: $L=2$, $x(p, w)$ is single valued. Walra's Law ✓

WARP ✓

1.1 Slutsky matrix is symmetric. hint [2.E.1 and 2E.2 or 2.F.3]

Since there are only 2 commodities, we will end up with a 2×2 Slutsky matrix, $S_{2 \times 2}$. To prove symmetry of S , we essentially just want to prove $S_{12} = S_{21}$

Firstly, Walra's law and WARP $\Rightarrow x(p, w)$ is homogenous degree of 0.

The (i, k) th entry of S is

$$S_{ik}(p, w) = \frac{\partial x_i(p, w)}{\partial p_k} + \frac{\partial x_i(p, w)}{\partial w} x_k(p, w)$$

$$S_{\ell k} \cdot P_k = \frac{\partial X_{\ell}(P, w)}{\partial P_k} \cdot P_k + \frac{\partial X_{\ell}(P, w)}{\partial w} X_k(P, w) \cdot P_k$$

Sum across k and apply result from 2.E.1, we obtain

$$\begin{aligned} \sum_k S_{\ell k} \cdot P_k &= \sum_k \frac{\partial X_{\ell}(P, w)}{\partial P_k} \cdot P_k + \frac{\partial X_{\ell}(P, w)}{\partial w} \sum_k X_k(P, w) P_k \\ &= \sum_k \frac{\partial X_{\ell}(P, w)}{\partial P_k} \cdot P_k + \frac{\partial X_{\ell}(P, w)}{\partial w} w = 0 \end{aligned}$$

$$S_{\ell 1} P_1 + S_{\ell 2} P_2 = 0 \quad \forall \ell = 1, 2$$

$$S_{11} P_1 + S_{12} P_2 = 0 \Rightarrow S_{12} = \frac{-S_{11} P_1}{P_2}$$

Next

$$P_e S_{ek}(P, w) = \frac{\partial X_e(P, w)}{\partial P_k} P_e + \frac{\partial X_e(P, w)}{\partial w} X_k(P, w) P_e$$

Sum over all good e and apply 2.E.3 and 2.E.2

$$\sum_e S_{ek}(P, w) P_e = \sum_e \frac{\partial X_e(P, w)}{\partial P_k} P_e + X_k(P, w) \sum_e \frac{\partial X_e(P, w)}{\partial w} P_e$$

$$= \underbrace{-X_k(P, w)}_{\textcircled{1}} + \underbrace{X_k(P, w)}_{\textcircled{2}} = 0 \quad \forall k=1, 2$$

① Comes from 2.E.2 $\sum_e P_e \frac{\partial X_e(P, w)}{\partial P_k} + X_k(P, w) = 0$

$$\sum_e P_e \frac{\partial X_e(P, w)}{\partial P_k} = -X_k(P, w)$$

② Comes from 2.E.3

$$\sum_e P_e \frac{\partial X_e(P, w)}{\partial w} = 1$$

Therefore $\sum_{\ell} S_{\ell k} P_{\ell} = 0 \quad \forall k = 1, 2$

$$S_{11} P_1 + S_{21} P_2 = 0$$

$$S_{21} = \frac{-S_{11} P_1}{P_2} = S_{12}$$

Therefore Slutsky matrix is symmetric for $L=2$

1. 2. It does not extend to the case where $L=3$

Recall $S P_{\textcircled{1}} = 0$ and $S' P_{\textcircled{2}} = 0$

$$S_{11} P_1 + S_{12} P_2 + S_{13} P_3 = 0 \quad \text{From } \textcircled{1} \quad \textcircled{3}$$

$$S_{11} P_1 + S_{21} P_2 + S_{31} P_3 = 0 \quad \text{From } \textcircled{2} \quad \textcircled{4}$$

$$\textcircled{3} - \textcircled{4} = (S_{12} - S_{21}) P_2 + (S_{13} - S_{31}) P_3 = 0$$

Since $P_2, P_3 > 0$, We may have the case that

$$S_{12} - S_{21} = P_3 \quad \text{and} \quad S_{13} - S_{31} = P_2$$

$$P_3 P_2 - P_2 P_3 = 0.$$

In this case $S_{12} \neq S_{21}$ and $S_{13} \neq S_{31}$

Therefore, we provided a counter-example

1.3 SARP is just a stronger version of WARP that forbids preference reversal. According to Prop. 2F2, if $x \in P(w)$ satisfies Walra's Law and WARP, then $S(P, w)$ is negative semi-definite. For $L=2$, we know it's symmetric, but not for the case when $L=3$.

$$2.1 \quad \begin{array}{l} x^2 p^1 = 15 < x^1 p^1 = 18 \\ x^1 p^2 = 15 > x^2 p^2 = 10 \end{array}$$

$$\begin{array}{l} x^1 \text{ R.P. to } x^2 \\ x^2 p^3 < p^3 x^3 \\ x^3 p^2 > x^2 p^2 \\ x^3 \text{ is R.P. } x^2 \end{array}$$

$$\begin{array}{l} x^1 p^3 = p^3 x^3 \\ x^3 p^1 > x^1 p^1 \end{array}$$

$$x^3 \text{ is R.P. } x^1$$

x^3 is R.P. x^1 is R.P. x^2 and x^3 is R.P. to x^2

there's no preference reversal, WARP U

2.2. From previous results, we already knew $x^3 \succeq x^2$. Now we to compare between x^4 and x^3 and x^4 and x^2 .

x^4 and x^3 :

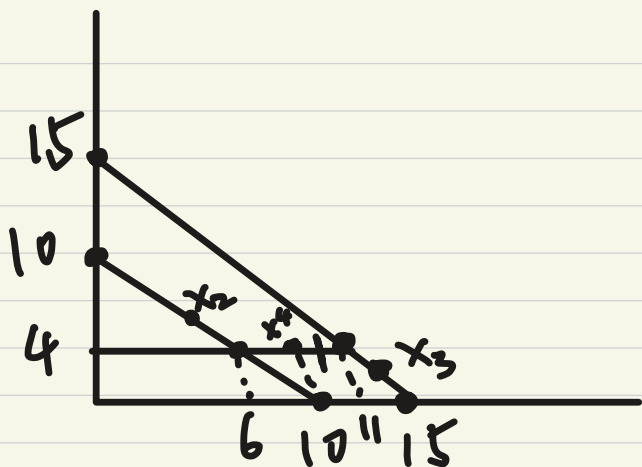
$$p^3 \cdot x^3 = 15 > p^3 \cdot x^4 = 14$$

then $x^3 \succeq x^4$.

By monotonicity, we may find $x^5 = x^4 + (0, 0.5)$ i.e. therefore $x^5 \succ x^4$

$$\text{but } x^5 \cdot p^3 = 4 \times 1 + 10.5 \times 1 = 14.5 < w^3 = 15$$

$$x^3 \succeq x^5 \succ x^4 \Rightarrow x^3 \succ x^4$$



$x^4 \cdot p^2 = 14 > w^2 = 10$
 It's inconclusive from WARP

Since $x^3 \succeq x^2$

then $x^t = \alpha x^3 + (1-\alpha)x^2 \succeq x^2$

Let first coordinate of x^t match with x^4 's.

$$\begin{aligned} \text{So } x^t &= 0.5x^2 + 0.5x^3 \\ &= (4, 8.5) \succeq x^2 \end{aligned}$$

But $x^4 \succ x^t$
 then $x^4 \succ x^2$

By transitivity $x^3 \succ x^4 \succ x^2$

With convexity only.

we may still conclude

$$x^3 \succeq x^4 \quad \text{due to WARP}$$

$$\text{we may still say } x^t = 2x^3 + (1-\alpha)x^4 \succeq x^4$$

But no further conclusion can be made without monotone assumption.

$$\text{For } x^4 \text{ and } x^2, \text{ we may still say } x^t = (4, 8.5) \succeq x^2$$

Even though $x^4 \succcurlyeq x^t$ and $x^4 \neq x^t$,

we still cannot conclude $x^4 \succ x^t \succeq x^2$.

It collapses.

2.3

	$p^1 = (1, 2)$	$p^2 = (1, 1)$	$p^3 = (1, 1)$	$p^4 = (2, 1)$
$x^1 = (12, 3)$	18	15	15	28
$x^2 = (5, 5)$	15	10	10	15
$x^3 = (3, 12)$	27	<u>15</u>	15	18
$x^4 = (4, 10)$	24	<u>14</u>	14	18

We still can obtain the previous result

$$x^1 \succeq x^2, \quad x^3 \succeq x^2 \text{ and } x^3 \succeq x^1$$

$x^4 p^3 \prec w^3$ but $x^3 p^4 = w^4$ WARP is violated.

check x^4 with x^3 and x^2

$$x^3 p^4 = x^4 p^4 \quad \text{WARP is violated.}$$

$$x^2 p^4 \prec x^4 p^4 = 18 \quad \text{it has to be}$$

$$x^4 p^2 > x^2 p^2$$

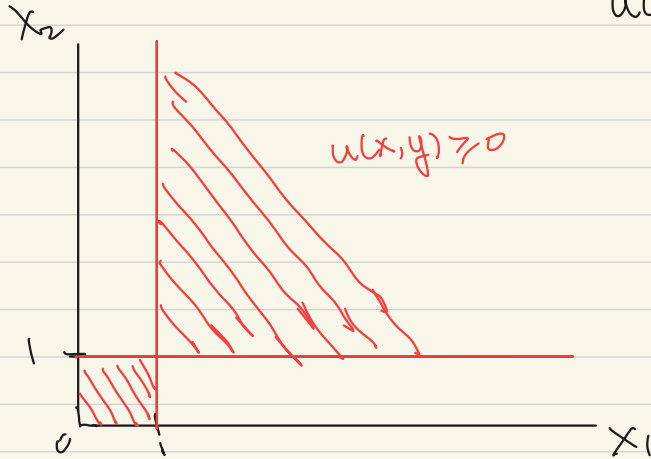
$$x^4 \succeq x^2$$

$$x^1 \succeq x^2, \quad x^3 \succeq x^2, \quad x^3 \succeq x^1 \quad \text{and} \quad x^4 \succeq x^2$$

Question 3.1

We assume standard non-negative space $\mathbb{R}_+^2$

$$u(x_1, x_2) = (x_1 - 1)(x_2 - 1)$$



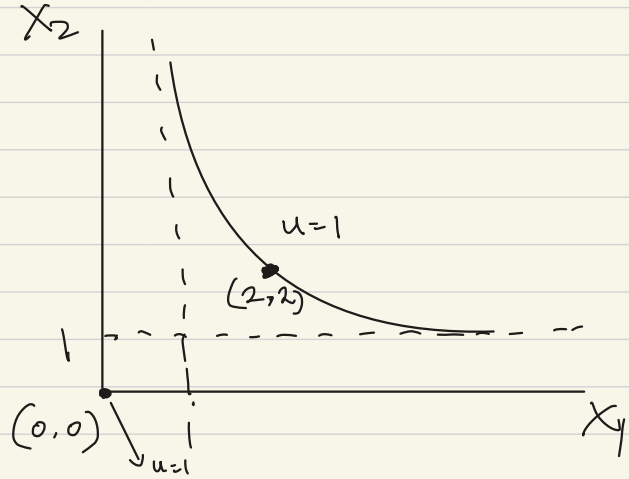
when $u=1 = (x_1-1)(x_2-1)$

$$x_2 - 1 = \frac{1}{x_1 - 1}$$

$$x_2 = \frac{1}{x_1 - 1} + 1 = \frac{1 + x_1 - 1}{x_1 - 1} = \frac{x_1}{x_1 - 1}$$

As $x_1 \rightarrow \infty$, $x_2 \rightarrow 1$

As $x_1 \rightarrow 1^+$, $x_2 \rightarrow \infty$



3.2 The preference represented by $u(x_1, x_2)$ is not convex. For a preference relation

to be convex, its upper contour sets $\{(x_1, x_2) : (x_1-1)(x_2-1) \geq k\}$ must be

convex as well. In specific example where $k=1$. We may consider

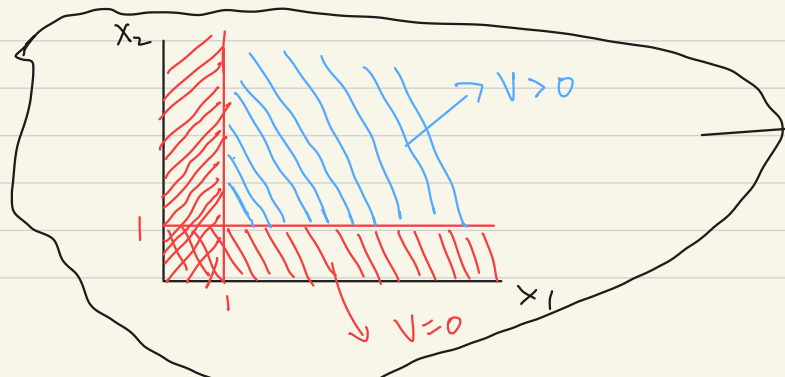
two points, i.e. $(0,0)$ and $(2,2)$. A linear combinations

$0.5 \cdot (0,0) + 0.5 \cdot (2,2) = (1,1)$ $u(1,1) = 0$, which has a lower utility level than $u(2,2) = 1$.

It's not locally non-satiated. In $x \in B_\varepsilon(0,0) \cap \mathbb{R}_+^2$, $\forall 0 < \varepsilon < 1$, we have $0 < u(x) < 1 = u(0,0)$, therefore it's not locally non-satiated. It can't be monotone if it's not locally non-satiated

3.3. If $x_1 > 1$ and $x_2 > 1$, then $v(x_1, x_2) = (x_1 - 1)(x_2 - 1)$

If $x_1 \leq 1$ or $x_2 \leq 1$, then $v(x_1, x_2) = 0$ and $x_1 \geq 0$ and $x_2 \geq 0$

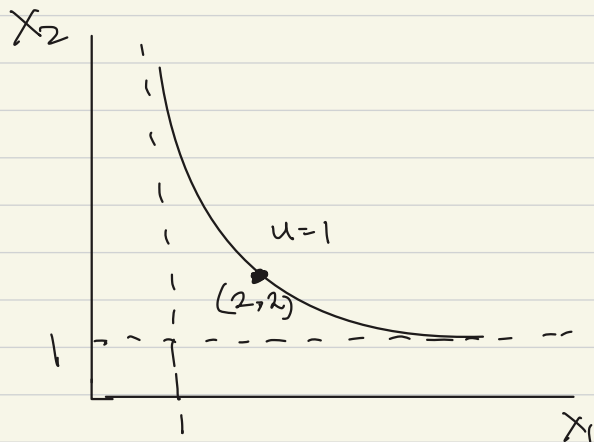


→ superior set $V \geq 0$
is the entire $\mathbb{R}_+^2$

When $V=1$, it must be the case that $x \gg 0$

$$x_2 = \frac{x_1}{x_1 - 1}$$

It's similar to the previous graph except for the inclusion of $(0, 0)$



3.4. The preference represented by x_1 and x_2 is convex. The upper contour set of $V \geq 0$ is the entire $\mathbb{R}_2^+$, which is convex.

Then $(x_1 - 1)(x_2 - 2) \geq k > 0$ if $x \gg 1$ covers entire $\{x \in \mathbb{R}_2^+ \mid x \gg 1\}$, which is a square area representing a convex set.

It's not non-locally satiated For $x \in B_\varepsilon(0,0) \cap \mathbb{R}_+^2$, $\forall 0 < \varepsilon < 1$, $U(x) = 0$
then it's not monotone.

3.5 Assume $P_1 + P_2 < w$, $(1,1)$ is affordable.
 $v(x_1, x_2)$ and $U(x_1, x_2)$. Same bundle x^* optimal for both?

Starting with v . Since we may still to increase $\$$ amount.

for x to obtain a higher utility in this case x_1 and $x_2 > 1$ while still ensure
the constraint bind. Since $v(x_1, x_2)$ strictly $\uparrow$ in both argu.

constraint bind we solve $\max v(x) = (x_1 - 1)(x_2 - 1)$
s.t. $x \gg 1$
and $P_1 x_1 + P_2 x_2 = w$

$$\mathcal{L}(x, \lambda) = (x_1 - 1)(x_2 - 1) - \lambda(w - P_1 x_1 - P_2 x_2)$$

$$\begin{aligned} L_{X_1} &= X_2 - 1 + \lambda P_1 = 0 \\ L_{X_2} &= X_1 - 1 + \lambda P_2 = 0 \end{aligned} \rightarrow \frac{X_2 - 1}{X_1 - 1} = \frac{P_1}{P_2} \quad \begin{aligned} (X_2 - 1)P_2 &= (X_1 - 1)P_1 \\ X_1 &= \frac{(X_2 - 1)P_2}{P_1} + 1 \end{aligned}$$

$$W = P_1 X_1 + P_2 X_2$$

$$P_1 X_1 = W - P_2 X_2$$

$$X_1 = \frac{W - P_2 X_2}{P_1}$$

$$\frac{W - P_2 X_2}{P_1} = \frac{(X_2 - 1)P_2}{P_1} + 1$$

$$W - P_2 X_2 = (X_2 - 1)P_2 + P_1$$

$$W - P_2 X_2 = P_2 X_2 - P_2 + P_1$$

$$2P_2 X_2 = W + P_2 - P_1$$

$$X_2 = \frac{W + P_2 - P_1}{2P_2}$$

$$X_1 = \frac{W + P_1 - P_2}{2P_1}$$

$$(X_1^* - 1)(X_2^* - 1) = \left(\frac{W + P_1 - P_2}{2P_1} - 1 \right) \left(\frac{W + P_2 - P_1}{2P_2} - 1 \right)$$

$$= \left(\frac{w - p_1 - p_2}{2p_1} \right) \left(\frac{w - p_1 - p_2}{2p_2} \right)$$

$$= \frac{(w - p_1 - p_2)^2}{4p_1 p_2} > 0$$

For u , it's different however. It reaches max at $(0,0)$ so $u=1$

Then another case would be when $x \gg 1$ we have $u(x^*) = \frac{(w - p_1 - p_2)p_2}{4p_1 p_2}$

Case when they don't share the same bundle

if $\left(\frac{w - p_1 - p_2}{4p_1 p_2} \right)^2 < 1$, we choose $(0,0)$

$$(w - p_1 - p_2)^2 < 4p_1p_2$$

$$w - p_1 - p_2 < 2\sqrt{p_1p_2}$$

$$p_1 + p_2 < w < 2\sqrt{p_1p_2} + p_1 + p_2$$

whenever this happens, we choose $(0,0)$

Otherwise if $x \gg 1$, then we are solving for the same optimization Prob for both utility with same optimal bundle